

Exp: 07

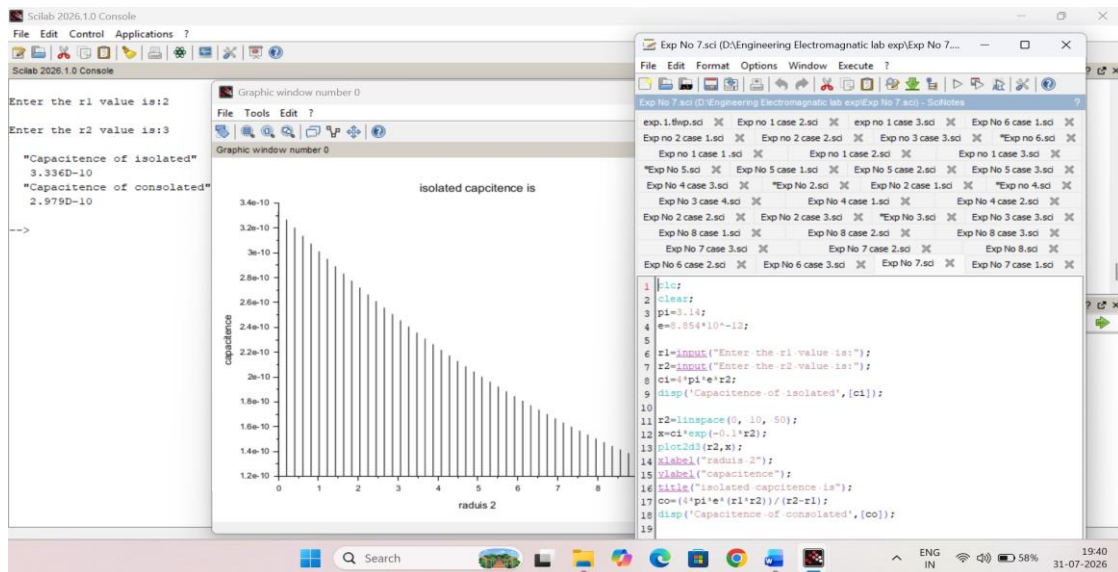
Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

Aim:

To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

Software required:

- SCILAB version 6.0.1



Exp No 307

$$r_1 = 2 \text{ cm}$$

$$r_2 = 3 \text{ cm}$$

Capacitance of Isolated

$$C_i = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{r_2}{r_1}\right)}$$

$$\ln(1.5) = 0.405$$

$$2\pi\epsilon_0 L = 2 \times 3.14 \times 8.85 \times 10^{-12}$$

$$\cancel{C_i = 3.365 \times 10^{-12}}$$

$$C = 3.365 \times 10^{-12}$$

Capacitance of Consolated

$$C_c = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{r_2}{r_1}\right)}$$

$$\ln\left(\frac{3}{2}\right) = 0.4055$$

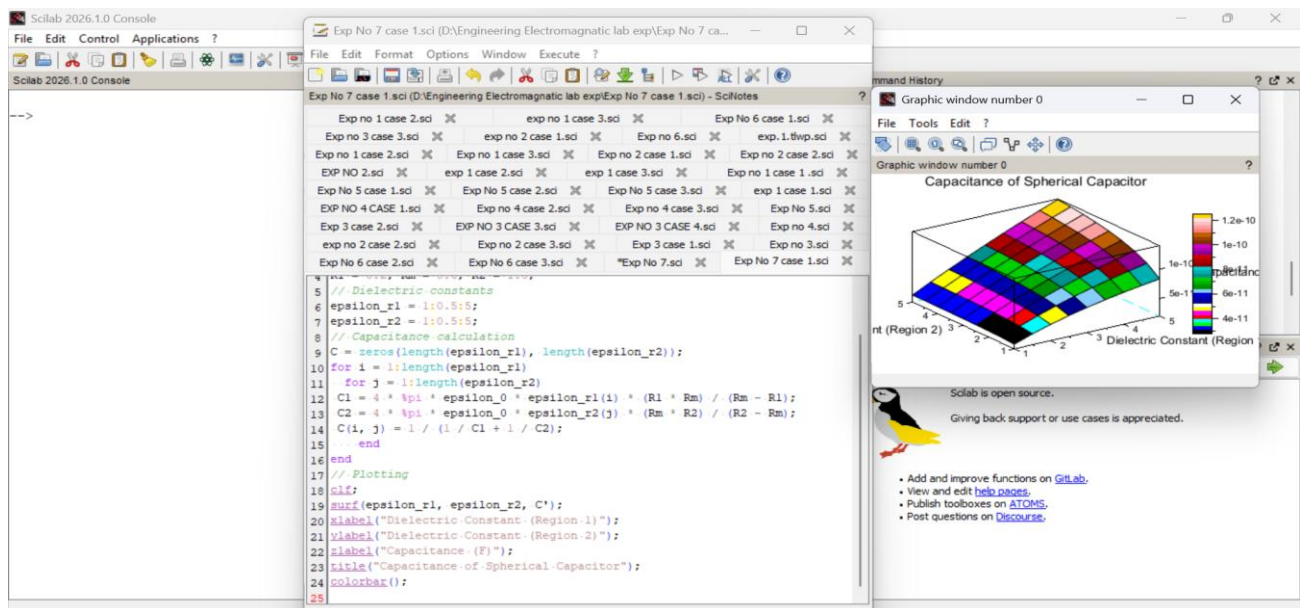
$$C = \frac{5.563 \times 10^{-12}}{0.4055} = 2.978 \times 10^{-12}$$

$$\cancel{C_c = 2.978 \times 10^{-12}}$$

$$C_c = 2.978 \times 10^{-12}$$

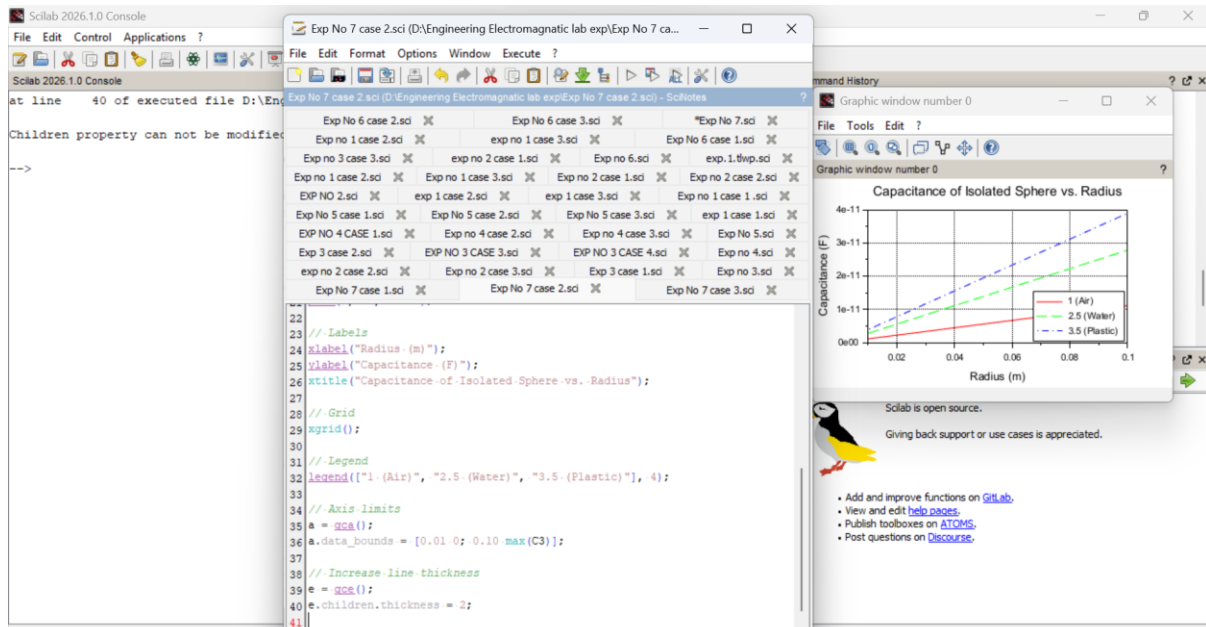
CASE 1 : -CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS

CODE: -



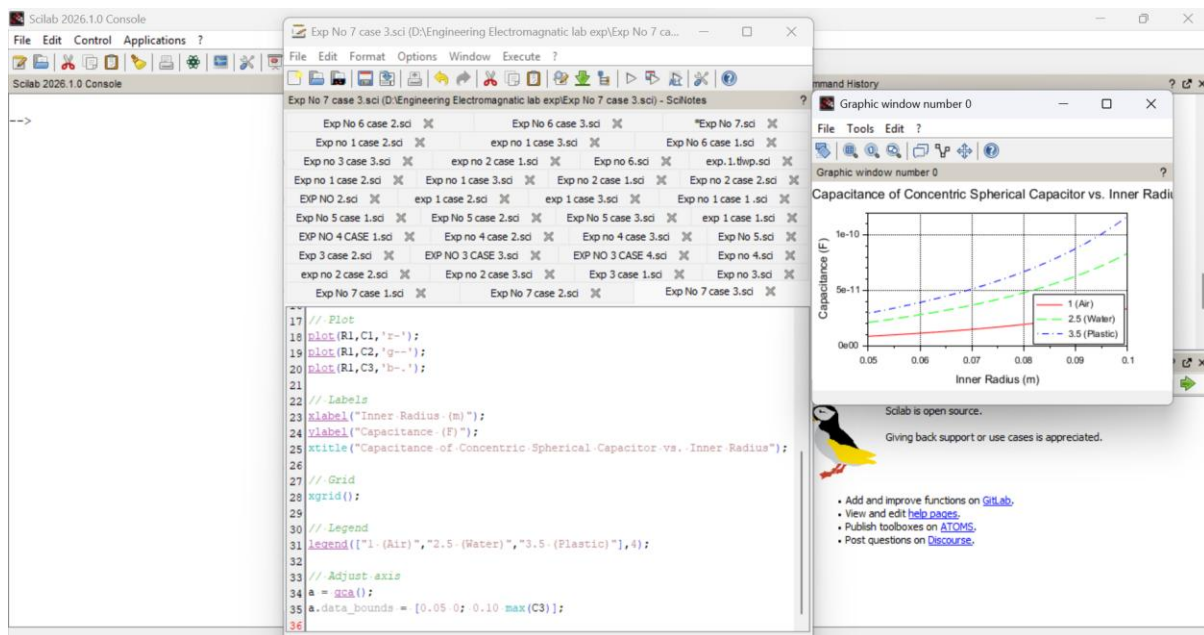
CASE 2: ISOLATED SPHERE

CODE:-



CASE 3: CONCENTRIC SPHERICAL CAPACITOR

CODE:-



Case 1: Concentric Spherical Capacitor with Different Dielectric Mediums

ϵ_{r1}	ϵ_{r2}	Capacitance (F)
1.0	1.0	2.73e-11
2.5	2.5	4.62e-11
5.0	5.0	7.12e-11

Case 2: Isolated Sphere

Radius (m)	Dielectric Constant (ϵ)	Capacitance (F)
0.01	1.0	1.11e-12
0.05	2.5	5.54e-12
0.10	3.5	12.33e-12

Case 3: Concentric Spherical Capacitor

Inner Radius (m)	Dielectric Constant (ϵ)	Capacitance (F)
0.05	1.0	1.13e-11
0.08	2.5	3.52e-11
0.10	3.5	6.11e-11

Result:

Thus, the program was verified for Capacitance of an isolated sphere and two concentric spheres using SCILAB was successfully.